

Lecture 12

October 4

Dependence

Learning objectives

- Distinguish the law of large numbers, which describes the stabilization of an average, from the central limit theorem, which describes its approximately normal sampling distribution under appropriate conditions.
- Derive the least-squares estimate for a regression through the origin, and contrast Legendre's squared-residual criterion with Laplace's minimum-variance justification under independent, mean-zero errors with equal variance.
- Distinguish estimating a common quantity from comparing measurements to detect a systematic effect, using Laplace's search for a lunar effect on barometric pressure.
- Use covariance terms to calculate the variances of sums and differences, and explain how accounting for correlated observations can change the relative precision and appropriate weighting of equations.
- Distinguish independence from zero covariance, and show why $U+V$ and $V-U$ are uncorrelated when U and V have equal variances without concluding that the sum and difference are independent.

CLT vs LLN

$X_1, \dots, X_n$ are independent random variables
 $E[X_i] = \mu, \text{Var}(X_i) = \sigma^2 < \infty$

$$P\left(\left|\frac{\sum X_i}{n} - \mu\right| > \epsilon\right) \rightarrow 0$$

Law of Large Numbers

Prob. the $\frac{\sum X_i}{n}$ is
very different from $E[X_i]$
is small.

Uncertainty around $\left(\frac{\sum X_i}{n} - \mu\right)$?

$$\sqrt{n}\left(\frac{\sum X_i}{n} - \mu\right) \rightarrow N(0, \sigma^2)$$

Central Limit Theorem

From lecture 11

lets try more Least Squares

We observe pairs $(x_1, y_1), \dots, (x_n, y_n)$ and want to relate them as

Find z such that $\sum (y_i - x_i z)^2$ is smallest

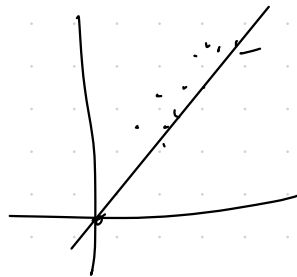
$$\min_z \sum (y_i^2 - 2x_i y_i z + x_i^2 z^2)$$

$$\frac{d}{dz} \sum (y_i^2 - 2x_i y_i z + x_i^2 z^2) = \sum -2x_i y_i + 2x_i^2 z$$

Set equal to zero

$$-2 \sum x_i y_i = -2z \sum x_i^2$$

$$z = \frac{\sum x_i y_i}{\sum x_i^2}$$



From Lecture 11

If errors are "just" mean 0 and Symmetric?

Write the same problem as $\underline{z_i = y_i - x_i z}$
Laplace knows sums!

maybe: $\sum m_i z_i = 0$

$$\sum m_i z_i = \sum m_i y_i - \sum m_i x_i z \quad \text{where } \cancel{m_i} \neq 0 \text{ weight}$$

$$0 = \sum m_i y_i - z \sum m_i x_i$$

$$\Rightarrow z = \frac{\sum m_i y_i}{\sum m_i x_i}$$

Not 0:

$$z = \frac{\sum m_i y_i}{\sum m_i x_i} + u$$

error term

$$u = \frac{\sum m_i z_i}{\sum m_i x_i}$$

average of errors

$$z = \frac{\sum x_i y_i}{\sum x_i^2} - \frac{\sum x_i y_i}{\sum x_i^2}$$

From Lecture 11

uses CLT to show $u \approx N(\text{mean}, \text{var})$

$$\underline{\text{var}(u)} = \text{var}\left(\frac{\sum m_i \varepsilon_i}{\sum m_i x_i}\right)$$

$$= \left(\frac{1}{\sum m_i x_i}\right)^2 \text{var}\left(\sum m_i \varepsilon_i\right)$$

$$= \frac{1}{(\sum m_i x_i)^2} \sum \text{var}(m_i \varepsilon_i) \quad \text{by independence}$$

$$= \frac{\sum m_i^2 \text{var}(\varepsilon_i)}{(\sum m_i x_i)^2} \leftarrow \text{the same}$$

$\propto \frac{\sum m_i^2}{(\sum m_i x_i)^2}$
proportional
to

To find z I need to
choose m_i
When you minimize $\frac{\sum m_i^2}{(\sum m_i x_i)^2}$
in m_i you'll find
 $m_i \propto x_i$

The conceptual change:

Legendre: Choose parameters that minimize squared residuals

Laplace: Choose a way to combine observations
that produces the most precise estimate.

Handout

$$Q(a) = (y_1 - a)^2 + (y_2 - 3a)^2$$

$$y_1 = z + \varepsilon_1$$

$$y_2 = 3z + \varepsilon_2$$

$$\varepsilon_1 = y_1 - z$$

$$\varepsilon_2 = y_2 - 3z$$

$$\sum \varepsilon_i^2 = \varepsilon_1^2 + \varepsilon_2^2$$

The errors of the two estimates

which one is the least squares rule?

$$A - z = \frac{\epsilon_1 + \epsilon_2}{4}, \quad \textcircled{B} - z = \frac{\epsilon_1 + 3\epsilon_2}{10}.$$

Each row has probability $1/4$. Complete the table.

ϵ_1	ϵ_2	$A - z$	$B - z$
-2	-2	-1	-0.8
-2	+2	0	0.4
+2	-2	0	-0.4
+2	+2	1	0.8

Variance and mean squared error

Both estimation errors have mean zero, so their variances equal their mean squared errors.

Square the four errors in each column and average.

$$\text{Var}(A) = \underline{0.5}, \quad \text{Var}(B) = \underline{0.4}.$$

Which rule has smaller mean squared error?

Is either error distribution normal?

Repeating the experiment

Perform **100 independent copies** of the two-reading experiment, always measuring the same z .

Average the 100 A estimates to obtain $\bar{A}_{100}$ and the 100 B estimates to obtain $\bar{B}_{100}$.

Imagine repeating this entire 100-pair study many times.

Which distributions remain two-point distributions?

Which should be approximately normal, and why?

$$\bar{A}_{100}, \bar{B}_{100}$$

because A, B are indep.

$$\sqrt{n}(\bar{A}_{100} - \mathbb{E}[A]) \sim N$$

have $\mathbb{E}[A], \mathbb{E}[B]$

$\text{var}(A), \text{var}(B) < \infty$

Variability after averaging

For independent repetitions, averaging r estimates divides their variance by r .

$$\text{Var}(\bar{A}_{100}) = \underline{0.005}, \quad \text{Var}(\bar{B}_{100}) = \underline{0.004}.$$

Which of the two approximately normal sampling distributions is narrower?

Bouvard

Laplace and Barometric Pressure

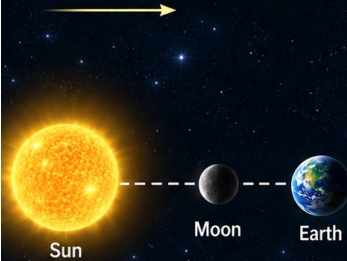
Why?

Tidal Theory

Syzygy: New Moon

The Sun, Moon, and Earth lie on one straight line.

Sunlight comes from the left.



Example New Moon

The Moon's illuminated side faces the Sun, so it appears dark from Earth.

Syzygy: Full Moon

Another syzygy arrangement: Earth lies between Sun and Moon.

Sunlight comes from the left.



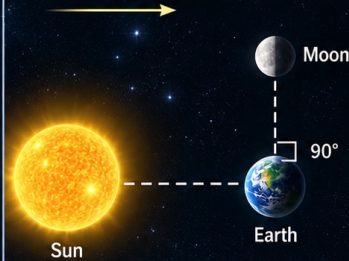
Example Full Moon

The Moon's illuminated side faces Earth, so it appears full.

Quadrature

The Moon is 90° from the Sun, as seen from Earth.

Sunlight comes from the left.

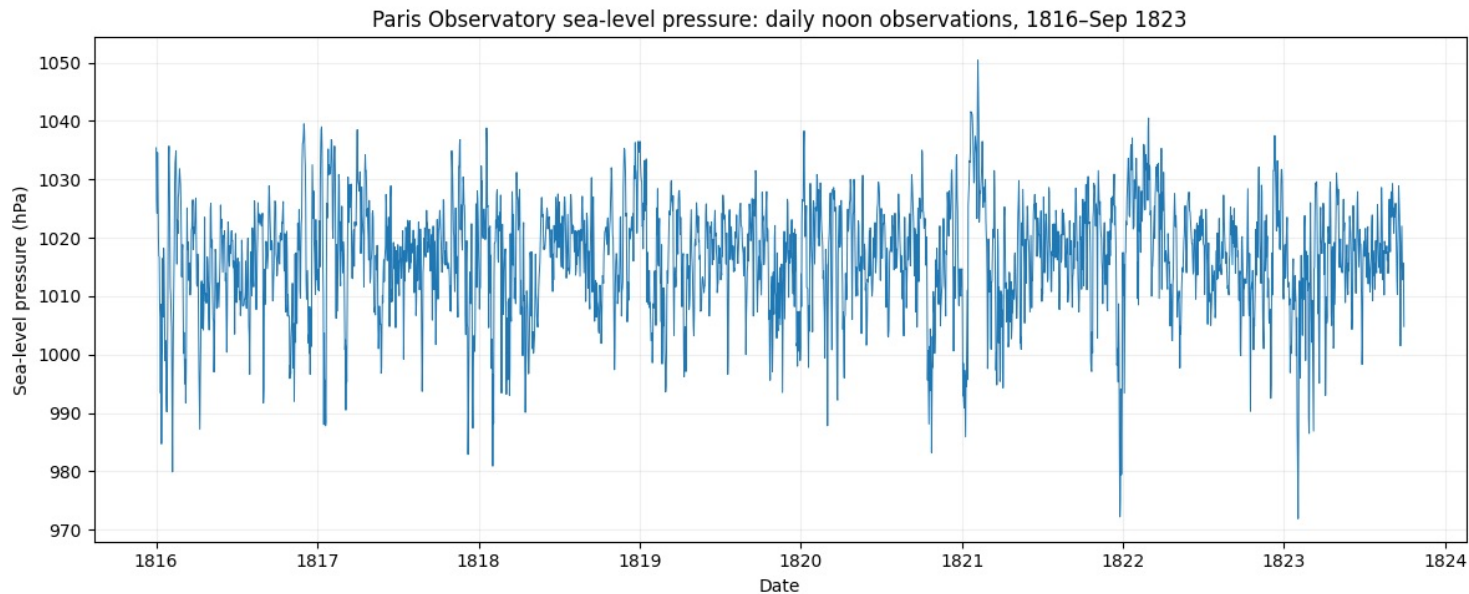


Example First or Third Quarter Moon

The Moon appears half illuminated (90° from the Sun).

Syzygy = straight-line alignment. Quadrature = 90° angle at Earth.

New moon and full moon are both syzygies; quarter moons are quadratures.



What can we say about these data?

Laplace's summary of the data

	9am	12 ^{pm}	3 ^{pm}
Syzygy day + <i>i</i>	A_i	A'_i	A''_i
Quadrature day + <i>i</i>	B_i	B'_i	B''_i

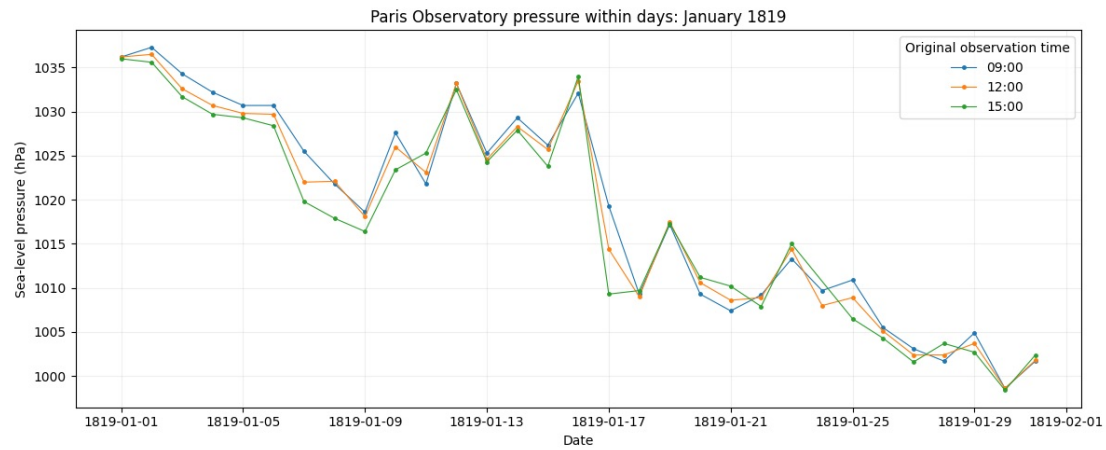
We write

$$E_i = (A''_i - A_i) - (B''_i - B_i)$$

difference between 3^{pm} and 9am

$$F_i = (2A'_i - (A_i + A''_i)) - (2B'_i - (B_i + B''_i))$$

difference between
noon and
average of
9am and 3^{pm}

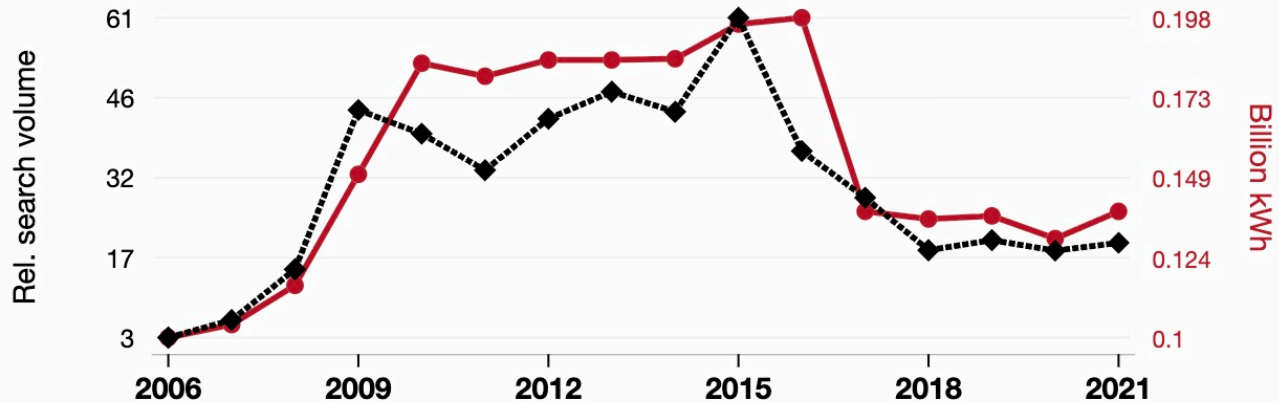


clear within day correlation

Google searches for 'Taylor Swift'

correlates with

Fossil fuel use in British Virgin Islands



◆--- Relative volume of Google searches for 'Taylor Swift' (Worldwide, without quotes) · Source: Google Trends

●— Total fossil fuel use in British Virgin Islands in billion kWh · Source: Energy Information Administration

2006-2021, $r=0.902$, $r^2=0.814$, $p<0.01$ · tylervigen.com/spurious/correlation/5825

Laplace proceeds with his physics model

$$x \cos(2ig) + y \sin(2ig) = E_i$$

$$y \cos(2ig) - x \sin(2ig) = F_i$$

}

g known

i represents shift from day

Are (E_i, F_i) independent?

Are (E_i, F_i) uncorrelated?

Laplace calculation

$$\text{Var}(E_i) = \text{Var}(A_i'' - A_i + B_i - B_i'')$$

$$= \text{Var}(A_i'') + \text{Var}(A_i) + \text{Var}(B_i) + \text{Var}(B_i'')$$

because the terms are independent, same variance

$$= \sigma^2 + \sigma^2 + \sigma^2 + \sigma^2 = 4\sigma^2$$

$$\text{Var}(F_i) = \text{Var}(2A_i' - (A_i + A_i'') - 2B_i' + (B_i + B_i''))$$

$$= \text{Var}(2A_i') + \text{Var}(A_i) + \text{Var}(A_i'') + \text{Var}(2B_i') + \text{Var}(B_i) + \text{Var}(B_i'')$$

$$= 4\sigma^2 + \sigma^2 + \sigma^2 + 4\sigma^2 + \sigma^2 + \sigma^2$$

$$= 12\sigma^2$$

$$\text{Var}(E_i) : \text{Var}(F_i)$$

$$1 : 3$$

But they aren't uncorrelated!

Let's look at $V_i = A_i' - A_i$ — first difference

$U_i = A_i'' - A_i'$ — second difference

$$E_i^{(A)} = U_i + V_i = A_i'' - A_i' + A_i' - A_i = A_i'' - A_i$$

$$F_i^{(A)} = V_i - U_i = A_i' - A_i + A_i' - A_i'' = 2A_i' - (A_i + A_i'')$$

Maybe $\text{var}(U_i) = \text{var}(V_i) = \sigma^2$

Maybe $\text{cov}(U_i, V_i) = \frac{1}{2} \sigma^2$

$$\begin{aligned} \Rightarrow \text{var}(U_i + V_i) &= 3\sigma^2 \\ &= \text{var}(U_i) + \text{var}(V_i) + 2\text{cov}(U_i, V_i) \\ &= \sigma^2 + \sigma^2 + 2\left(\frac{1}{2}\sigma^2\right) \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{var}(V_i - U_i) &= \sigma^2 \\ &= \text{var}(V_i) + \text{var}(U_i) - 2\text{cov}(U_i, V_i) \\ &= \sigma^2 + \sigma^2 - 2\left(\frac{1}{2}\sigma^2\right) \end{aligned}$$

$$\begin{array}{ccc} \text{var}(E_i) & \text{var}(F_i) \\ 3 & : & 1 \end{array}$$

Laplace's mistake

- He weighted the equations proportional to the precision
- His weights $1:3$
- Correct weights $3:1$
 - Mistake $\Rightarrow$ 9 fold!

Independence and correlation

Are E_i and F_i correlated?

$$\begin{aligned}\text{Cov}(V_i + u_i, V_i - u_i) &= \text{Cov}(V_i, V_i) + \underbrace{\text{Cov}(u_i, V_i)}_{=0} - \underbrace{\text{Cov}(V_i, u_i)}_{=0} - \text{Cov}(u_i, u_i) \\ &= \text{Var}(V_i) - \text{Var}(u_i) \\ &= \sigma^2 - \sigma^2 \\ &= 0\end{aligned}$$

So Laplace got lucky on the correlation
between E_i , F_i --